

Research Article

Antibiotic Susceptibility Pattern of Canine Coagulase Positive and Coagulase Negative *Staphylococcus* spp. in a Hot and Dry Region of India

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Keywords:

Staphylococcus
antibiotic resistance
MRSA
MRCoNS
canine

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The emergence of antimicrobial resistance among *Staphylococcus* spp. isolated from clinical cases of canines should be continuously monitored hence the present study was formulated to ascertain the antibiotype and methicillin resistance in coagulase positive and coagulase negative staphylococci of canine skin and associated mucous membrane affections from a hot and dry region of India. A total of 165 clinical samples were collected and *Staphylococcus aureus* was identified by conventional bacteriological methods and PCR. Antibiotic susceptibility test was done against commercially available antibiotic impregnated discs as per Clinical and Laboratory Standards Institute (CLSI) method. Methicillin resistance was determined by plate methods and then via PCR of *mecA* gene. These 165 samples yielded, 88 (53.33%) isolates of genus *Staphylococcus* and 46 *S. aureus* and 51/88 (57.95%) isolates were coagulase positive staphylococci. Total 55 (62.5%) isolates showed susceptibility to Ceftriaxone/Sulbactam, 37 (42.05%) to Ciprofloxacin, 26 (29.55%) to Oxacillin, 24 (27.27%) to Penicillin, and 10 (11.36%) to Gentamicin. Total 21 methicillin-resistant *S. aureus* (MRSA) and 12 methicillin-resistant coagulase negative staphylococci (MRCoNS) were found on phenotypic basis whereas the *mecA* gene was detected in 6/21 MRSA and 2/12 MRCoNS isolates. *Staphylococcus* spp. showed increased level of resistance against commonly used antibiotics. The higher prevalence of methicillin resistance found with phenotypic methods than to *mecA* PCR indicates toward additional mechanisms responsible for emergence of MRS, especially in CoNS.

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Introduction

Staphylococcus spp. is one of the most significant pathogens, involved in wide variety of canine infections. Clinical microbiologists consider coagulase production as an important identification and virulence factor and these isolates are designated as coagulase-positive staphylococci (CoPS).¹ CoPS viz. *Staphylococcus aureus*, *S. intermedius*, *S. pseudintermedius*, *S. delphini*, and *S. schleiferi* subspp *coagulans* together constitute the most frequent causes of canine staphylococcal infections. Coagulase-negative staphylococci (CoNS) are thought to be less pathogenic; however, their involvement in the infections of dogs has also been widely confirmed.² The 2 groups are characterized by alarmingly increasing rate of antibiotic resistance and now gained notoriety to be resistant against virtually any antibiotic drug. In particular, the issue of resistance is important in the case of multidrug-resistant (MDR) and methicillin-resistant (MR) strains.³ Methicillin-resistant *S. aureus* (MRSA) are characterized by the possession of staphylococcal cassette chromosome *mec* (SCC*mec*) element having the *mecA* gene, a penicillin-binding protein conferring resistance to practically all β -lactams with the exception of ceftaroline. In addition, MRSA often carry additional resistance traits for other antibiotics and limit the options for therapy.⁴ Determination of antimicrobial susceptibility of clinical isolates is required for choosing most suitable therapeutic antibiotic and for monitoring the intra species or inter species spread of resistant strains throughout the populations. These susceptibility data for a large number of isolates in an area or region can also be useful

to practicing clinicians including veterinarians.⁵ Until now antibiotic susceptibility tests were carried out on phenotypic basis by disc diffusion and broth dilution methods, which are time consuming methods. But today, PCR based rapid and convenient detection of antibiotic resistance have become feasible because of availability of antibiotic resistance genes sequence data.⁶ Application of this technology at small town hospitals and clinics may prove highly useful in curbing the global menace of increasing antibiotic resistance. Hence, due to paucity of reports on animal isolates than to human MRS strains⁶ and approximation of canine, human (and bovines) habitats in India, the present study was intended for determination of presence and antibiotic susceptibility pattern of *Staphylococcus* spp., notably of *S. aureus* upon skin and associated membrane affections of canines in a hot and dry region of India with special emphasis on methicillin resistance in CoPS and CoNS.

Materials and Methods

Sample Collection and Primary Processing

The work was carried out in the Department of Veterinary Microbiology, College of Veterinary Science and Animal Husbandry, Sardarkrushinagar Dantiwada Agricultural University, Sardarkrushinagar, Banaskantha, Gujarat (24.3455° N, 71.7622° E). A total of 165 clinical samples were collected from various clinical affections of dog during the time period of 2016-17 and processed for the presence of *Staphylococcus* spp. as per standard bacteriological techniques.^{7,8} Tube coagulase test, Vogus Proskauer test, mannitol and maltose fermentation test were specifically used to differentiate the related species. *S. aureus* was identified on the basis of PCR using primer pairs described previously as 16S rDNA and 442sa primer by Lovseth et al⁹ and Martineau et al,¹⁰ respectively.

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Phenotypic In Vitro Antibiotic Susceptibility Test

All the *Staphylococcus* spp. isolates obtained were subjected to in vitro antibiotic susceptibility test, as per the method described in CLSI manual.¹¹ Briefly, after overnight culture in 37°C BHI broth, each isolate (0.5 McFarland standard) was swabbed on Muller Hinton agar plates (Hi Media Pvt. Ltd., Mumbai). Antibiotic discs (Hi Media, Mumbai) of antibiotics viz., Penicillin G (10 Units), Ciprofloxacin (30 mcg), Gentamicin (10 mcg), Ceftriaxone/Sulbactam (30/15 mcg) and Oxacillin (5 mcg) were then placed on the agar surface and incubated overnight aerobically at 37°C. Zones of inhibition were measured and compared with zone size interpretative table furnished by the manufacturer and graded as either sensitive or resistant.

Phenotypic Characterization of Methicillin Resistance

A bacterial suspension of each isolate was made and turbidity was adjusted to 0.5 McFarland and swabbed on Mueller-Hinton agar containing 4.0% of NaCl and 6 µg/ml of oxacillin (Hi-Media, Mumbai). Plates were incubated at 35°C for 24 hrs. The strain showing growth on the plate containing oxacillin was considered to be resistant to methicillin. A swab was dipped in the suspension and squeezed on agar surface and streaked upon a CHROMagar plate for MRSA (Hi-Media, Mumbai, India). The medium contains peptone (35 gm/lit), NaCl (17.5 gm/lit), proprietary inhibitory and chromogenic substance and it was supplemented with cefoxitin solution (Hi-Media, Mumbai, India). The growth of any pink to mauve colony was considered to be positive, indicating of MRSA.

Molecular Characterization of Methicillin-Resistant Isolates

It was carried out in PCR test using *mecA* gene-specific primers for methicillin resistance as per method described by Strommenger et al.¹² The *mecC* PCR of *mecA* negative isolates was carried out with primer pair (5'-GAA AAA AAG GCT TAG AAC GCC TC-3' and 5' GAA GAT CTT TTC CGT TTT CAG C-3') as described previously.¹³ PCR mixture was prepared by using PCR master mix (Qiagen, Germany) in final reaction volume of 25 µl in a thin walled 200 µl PCR tubes (Genexy Ltd, India) and reaction was carried out for 35 cycles using a Nexus Mastercycler (Eppendorf). The details about primers employed, cycling conditions and desired amplicon of all the PCR tests in the study have been depicted (Table 1). Amplified products (5

µl) was loaded on to 1.5 per cent agarose gel (Amresco India), electrophoresed and documented with Gel document system (Thermo Scientific, India).

Results

Isolation and Prevalence of *S. Aureus*

A total of 165 samples collected yielded, 88 (53.33%) isolates of *Staphylococcus* spp. on the basis of bacteriological (phenotypic) and genus specific 16S rDNA based PCR methods. Clinically, 19 (21.59%), 28 (31.82%), 12 (13.64%), 15 (17.04%) and 14 (15.91%) isolates were recovered from abscess/wound, pyoderma, respiratory problems, eye infections and otitis, respectively (Table 2). *S. aureus* was genotypically confirmed by obtaining a 442 bp amplicon in *sa442* gene-specific PCR in 46 out of 88 (52.27%) isolates. The presence of *S. aureus* in such affections was found as 27.87% (46/165). In tube coagulase test, 51/88 (57.95%) isolates were found positive for coagulase enzyme, but surprisingly, 5 isolates which were detected positive in tube coagulase test, were found negative in *sa442* gene PCR.

Antibiotic Susceptibility Test

Out of 88 isolates of *Staphylococcus* spp., 55 (62.5%) showed susceptibility to ceftriaxone/sulbactam, 37 (42.05%) to ciprofloxacin, 24 (27.27%) to penicillin, and 10 (11.36%) to gentamicin. Only, 26 (29.55%) isolates were susceptible to oxacillin, therefore, the other resistant isolates were considered as methicillin-resistant bacteria.

Phenotypic and Genotypic Characterization of Methicillin Resistance

Out of 46 PCR positive *S. aureus* isolates, 23 isolates showed growth on the MH plate containing oxacillin, which were considered as MRSA. Whereas out of 42 other *Staphylococcus* spp. isolates, 10 isolates were found as methicillin-resistant coagulase negative staphylococci (MRCoNS). On the same line, Chromagar plate method detected, 21/46 and 12/42 isolates as MRSA and MRCoNS. The MRCoNS were isolated from canine dermatitis (6), pus samples (3) and nasal swab (3) samples. The isolates, which showed phenotypic resistance to methicillin, were subjected to PCR for detection of *mecA* gene, targeting a 533 bp fragment. Out of 23 MRSA isolates, 6 isolates (36-SDAU, 48-SDAU, 70-SDAU, 71-SDAU, 101-SDAU and 135-SDAU) were found positive.

Table 1

Detail of Sample Collection and Percent Positivity for *Staphylococcus* spp

Types of Sample Collected	No. of Sample Collected	Positive for <i>Staphylococcus</i> spp.	Percent(N = 88)
Pus/Exudates (abscess/wound)	45	19	21.59%
Skin scrapings	52	28	31.82%
Nasal swabs	27	12	13.64%
Eye (ocular) swabs	22	15	17.04%
Otitis externa (Ear swabs)	19	14	15.91%
Total	165	88	100

Table 2

Details About Cycling Conditions and Amplicon Size of PCR Protocols of the Study

Primers	Cycling Conditions					Amplicon
	Initial Denaturation	Denaturation	Annealing	Extension	Final Extension	
16s-rDNA	94°C 5 min.	94°C 30 sec.	60°C 30 sec.	72°C 60 sec.	72°C 5 min.	228 bp
sa442	94°C 5 min.	94°C 30 sec.	55°C 30 sec.	72°C 1 min.	72°C 7 min.	108 bp
<i>mecA</i> gene	94°C 4 min.	94°C 30 sec.	55°C 30 sec.	72°C 1 min.	72°C 4 min.	533 bp

Table. 3Characteristics of *mec*APCR-Confirmed MRSA and MRCoNS Isolates From Various Clinical Conditions

Attributes/Isolates	36-SDAU	48-SDAU	70-SDAU	71-SDAU	101-SDAU	135-SDAU	30-SDAU	158-SDAU
Clinical condition	Pyoderma	Abscess on skin	Pyoderma	Pyoderma	Skin infection	Wound	Pyoderma	Skin infection
Colony pigment on Nutrient Agar	Golden yellow	Golden yellow	Golden yellow	Golden yellow	Golden yellow	Golden yellow	White	white
Mannitol fermentation on MSA	Positive for mannitol fermentation							
Gram's staining	Gram +Ve Cocci in bunch							
Biochemical tests								
Catalase	Positive for catalase							
Oxidase	Negative for oxidase							
Phosphate test	Positive for phosphate test							
Maltose fermentation	Positive maltose fermentation test							
Coagulase test	+ve	+ve	+ve	+ve	+ve	+ve	-ve	-ve
Hemolysin production	Beta	Beta	Alpha	Alpha	Gamma	Beta	Alpha-Beta	Gamma
16S-rDNA-based PCR	+ve	+ve	+ve	+ve	+ve	+ve	+ve	+ve
(Genus specific)								
sa442 gene-based PCR	+ve	+ve	+ve	+ve	+ve	+ve	+ve	+ve
(Species specific)								
Coa gene-based PCR	+ve	+ve	+ve	+ve	+ve	+ve	-ve	-ve
Spa gene-based PCR	+ve	+ve	+ve	+ve	+ve	+ve	-ve	-ve
Phenotypic identification of MRSA								
Oxacillin screen agar test	+ve	+ve	+ve	+ve	+ve	+ve	+ve	+ve
CHROM agar	+ve	+ve	+ve	+ve	+ve	+ve	+ve	+ve
Disk diffusion	+ve	+ve	+ve	+ve	+ve	+ve	+ve	+ve
Genotypic identification								
<i>mecA</i> gene-based PCR	+ve	+ve	+ve	+ve	+ve	+ve	+ve MRCoNS	+ve MRCoNS
	MRSA	MRSA	MRSA	MRSA	MRSA	MRSA		
Sequencing		MRSA-specific sequence	MRSA-specific sequence	MRSA-specific sequence	MRSA-specific sequence			

Among 12 MRCoNS isolates, presence of *mecA* gene confirmed resistance to methicillin in 2 isolates (30-SDAU and 158-SDAU) (Table 3). None of *mecA* PCR negative isolates was found positive in *mecC* PCR.

Discussion

In comparison to our findings, very few studies recorded a very high prevalence of genus *Staphylococcus*,¹⁴ but equating prevalence of *S. aureus* were reported as 24%,¹⁵ 25%,¹⁶ and 29.9%¹⁷ from India, its neighboring nation Bangladesh and Europe, respectively. In recent reports, slightly higher finding as 42%¹⁸ and 50%^{18,20} have also been reported, whereas lower prevalence as 12.7% has also been reported by Tarazi et al.⁴ from Jordan and Katarzyna et al.²¹ from Poland, respectively. The difference recorded might be due to predominance of another CoPS, *Staphylococcus pseudintermedius*, which was found as important pathogen solely²² or with *S. aureus*¹⁵ in canine skin and associated infections.

The present study indicates very high level of antibiotic resistance and none of isolate was found as pan sensitive against all types of tested antibiotics. Higher level of resistance against broad spectrum aminoglycoside gentamicin was also found by other researchers.^{16,23,24} Though contrastingly, MSSA showed 92% susceptibility against this class of antibiotics.¹⁸ The susceptibility of MSSA and resistance of MRSA against similar class of antibiotics¹⁹ indicate association of MR resistance with resistance to other classes of antibiotics also as described by Tarzai et al.⁴

Interestingly, better susceptibility toward penicillin in the present work was contrasting to 81.7%¹⁸ and 100.0%^{16,22} of penicillin resistant isolates documented in the recent literature and showed revival of penicillin susceptibility toward bacteria. Here, susceptibility toward combination antibiotics ceftriaxone/sulbactam was found effective than single beta lactam drug. Similar level of susceptibility to amoxicillin-clavulanic acid in *Staphylococcus* spp. isolates from dogs was reported by Reddy et al.¹⁴ by combining both MRSA and MSSA strains. But contrary to this report, 93.51%²⁵ and 89.55%²⁶ of *Staphylococcus* spp. isolates from dogs were found susceptible against ceftriaxone, respectively. Higher susceptibility toward combination disc than single beta lactam antibiotic clearly showing that now more number of bacterial isolates are possessing genes of beta lactamases enzyme production. Where sulbactam and clavulanic acid block the enzyme and this synergistic effect of antibiotics is well documented.

Heterogeneous MRS populations with variable *mecA* expression were better detected by Chromagar with cefoxitin than medium with oxacillin alone, which is a weak inducer of PBP2a production. Though, strong correlation between the *mecA* gene PCR and phenotypic methicillin resistance has also been reported previously^{27,28} but this was in sharp contrast to our results. This may be due to the fact that accurate detection of methicillin resistance in *Staphylococcus* spp. by phenotypic methods becomes difficult, due to phenomenon of heterogeneous resistance against methicillin and oxacillin.²⁹ Therefore, results of PCR of *mecA* was required to correctly identify the most heterogeneous and borderline strains as indicated by earlier workers.^{30,31}

The prevalence of *mecA* gene among all *Staphylococcus* spp. and *S. aureus* of canine origin was found as 8/88 (09.09%) and 6/46 (13.04%), respectively. Our results of *mecA* gene prevalence corroborated to similar work reported from Bangladesh as 11.11%,¹⁶ but found higher than reports from Jordan,⁴ Bangladesh,¹⁸ Indonesia,²⁰ and Europe.¹⁵ Though much higher presence of *mecA* gene, i.e., 23% was found from India by Yadav et al.¹⁸ in canine and other domestic animals. The other important feature of study was the presence of methicillin resistance in CoNS. Though these have been reported earlier^{2,32,33} from different countries, but simply as nasal carriers,^{2,34} whereas in the present work, these were isolated from clinical affections, as previously reported by Kern et al.³⁴

Interestingly, despite the higher detection limits of PCR over classical bacteriological methods, there was one isolate which was positive in *mecA* gene PCR but negative in both phenotypic detection tests, possibly target gene was present but not expressed. These findings also provide evidence that there are addition mechanisms responsible for beta-lactam resistance of MRSA other than the presence of *mecA* gene. Most notable among them is beta-lactam resistance gene *mecC* (formerly *mecALGA251*),¹³ present on SCCmec, though presence of such gene was not detected through PCR in the present work.

Author Contribution

Sejalben S. Chaudhari: As master's student she carried out sampling and most of wet lab experiments; Harshadkumar C. Chauhan: As major guide of student, he planned the study and supervise the overall work; Kishan K. Sharma: Major data interpretation and manuscript preparation; Sandip S. Patel: Identification of isolates and phenotypic AST; Arun C. Patel: AST and PCR; Sushil K. Mohapatra: PCR and sequencing; Mehulkumar D. Srimali: Sample collection and isolation and identification; Bharatsingh Chandel: Overall supervision.

Funding

None.

Ethical Approval

Not required.

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